Question Number- 35

Problem Statement:

Code Summarization Using AST and NLP

❖ **Objective**: Generate human-readable summaries of programs for audit purposes.

❖ **Compiler Concepts**: AST traversal, semantic analysis, code representation

❖ **Security Relevance**: Simplifies security audits and code reviews.

❖ **Expected Work**

• Extract structural AST patterns

• Train text-generation model

• Summarize program behavior

❖ **Deliverables**:AST-based code summarization tool

Execution Plan and Report

Phase 1: Problem Understanding & Basic Foundations

Week 1 – Problem Definition & Context Understanding
Activities
• Understand the assigned problem statement
• Identify application domain and scope
• Define objectives, constraints, and expected outcomes
Deliverables
• Problem definition document
• Problem scope & objectives
• Initial understanding report

Week 2 – Literature Survey & Gap Identification
Activities
• Review 10–15 recent standard papers, standards, or tools
• Study existing approaches and limitations
• Identify research gaps and motivation
Deliverables
• Literature review table
• Comparative analysis
• Problem gap statement

Phase 2: Requirement Analysis & System Design

Week 3 – Requirements & Threat Analysis
Activities
• Identify functional and non-functional requirements
• Define security, performance, and scalability requirements
• Perform threat or risk analysis (if applicable)
Deliverables
• Software Requirement Specification (SRS)
• Threat model/risk analysis document

Week 4 – Architecture & Design PlanningActivities
• Design overall system architecture
• Identify modules and interactions
• Select tools, frameworks, datasets, and platforms
Deliverables
• Architecture diagram
• Module interaction diagram
• Technology stack justification

Phase 3: Core Development Phase

Week 5 – Language / Model / Interface Design
Activities
• Define grammar, data models, or interface schemas
• Design APIs or DSLs (if applicable)
• Establish input/output formats
Deliverables
• Grammar specification or data model
• Sample inputs and outputs

Week 6 – Frontend / Parsing / Data Ingestion
Activities
• Implement input parsing or data ingestion
• Build syntactic and basic semantic validation
• Handle error detection and reporting
Deliverables
• Parser or data ingestion module
• Syntax validation results

Week 7 – Core Logic Implementation
Activities
• Implement core algorithms or compiler passes
• Build analysis or transformation modules
• Integrate rule-based or AI components
Deliverables• Functional core engine
• Intermediate outputs and logs

Week 8 – Intermediate Representation & Analysis
Activities

- Designed neural intermediate representation using transformer embeddings
- Implemented semantic analysis through attention mechanisms and vector operations
- Performed pattern detection via embedding similarity in 768-dimensional space

Deliverables

- IR Design Documentation: Neural IR specification using CodeT5 embeddings (token vectors → hidden states → semantic representations)
- Analysis Results: ml_summarizer.py module producing semantic summaries, pattern detection outputs, and analysis reports

Phase 4: Intelligence, Optimization & Security

Week 9 – Advanced Logic / AI Integration
Activities
• Integrate ML/AI models or heuristics

• Train or fine-tune models if applicable
Deliverables
• Trained model
• Evaluation metrics

Week 10 – Optimization & Security Enforcement
Activities
• Optimize performance, memory, or energy
• Enforce security rules or constraints
Deliverables
• Optimized implementation
• Security analysis report

Phase 5: Evaluation & Validation

Week 11 – Testing & Validation
Activities
• Functional testing
• Stress testing and corner cases• Security or correctness testing
Deliverables
• Test cases and results
• Bug-fix documentation

Week 12 – Evaluation & Benchmarking
Activities
• Compare with baseline systems
• Measure accuracy, efficiency, scalability
Deliverables
• Comparative performance report
• Graphs and metrics
Phase 6: Explainability, Documentation & Refinement

Week 13 – Explainability & Interpretability
Activities
• Generate explanations or visualizations
• Interpret system behavior
Deliverables
• Explanation reports
• Visualization artifacts

Week 14 – Final Refinement & Integration
Activities
• Refine codebase
• Integrate all modules
• Final testing
Deliverables
• Stable system build
• Integration report

Phase 7: Reporting & Evaluation

Week 15 – Documentation & PresentationActivities
- Prepare final report and documentation
- Create presentation slides

Deliverables
- Report
- PPT for evaluation

Week 16 – Demonstration & Assessment
Activities
- Live demonstration
- Viva voce / evaluation
- Submission of code and documents

Deliverables
- Final source code
- Demo recording